Red Light and Speed Cameras Deployment and Traffic Crash Outcomes in Chicago

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3.3.2026

Agenda

I. Overview

II. Spatial Distribution of Transportation Camera Deployment in Chicago

III. Event Study

IV. Dashboard

V. Connection to Policy

Motivation & Research Question

Motivation:

Traffic accidents affect public safety and economic costs. Examining how the deployment of red light cameras and speed cameras relates to crash patterns is important for evaluating traffic safety policy design and resource allocation.

Research Question:

1. Are Cameras Placed in High-Risk Areas?
2. Do Cameras Reduce Crashes or Severity?

Method

Data:

All datasets used in this project are sourced from the Chicago Data Portal (https://data.cityofchicago.org).

We used 4 datasets in total:

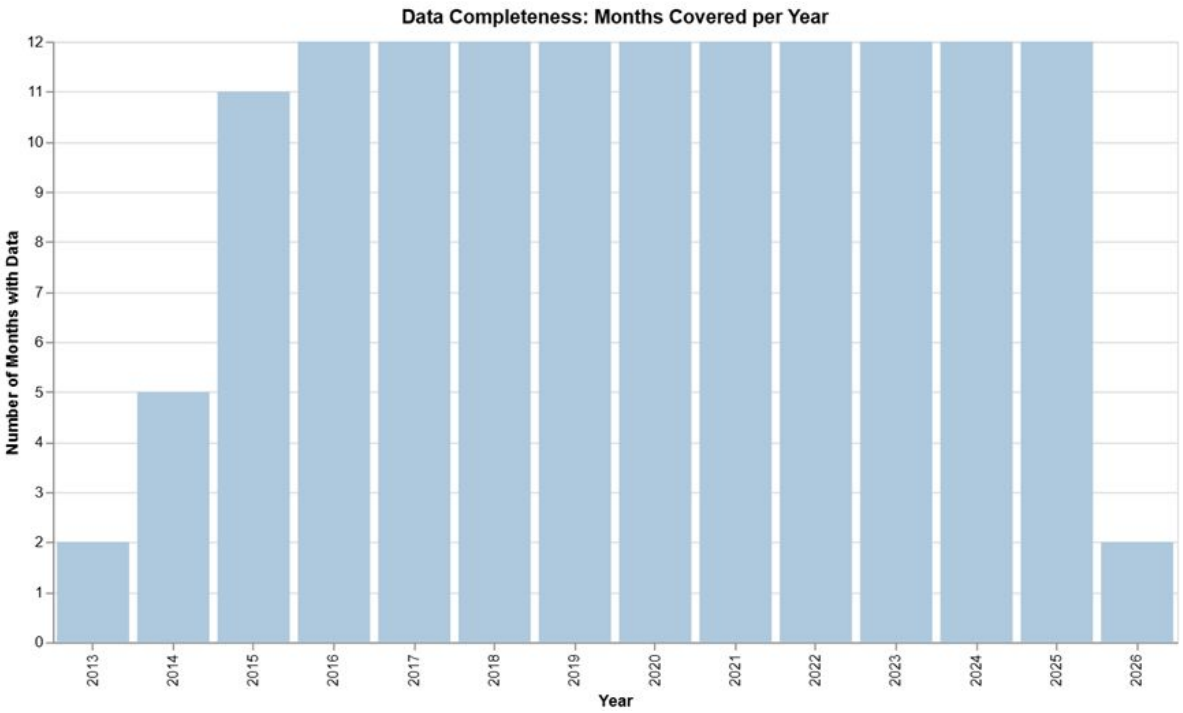
* Neighborhood boundaries in Chicago
* Traffic Crashes
* Speed Camera Locations
* Red Light Camera Locations

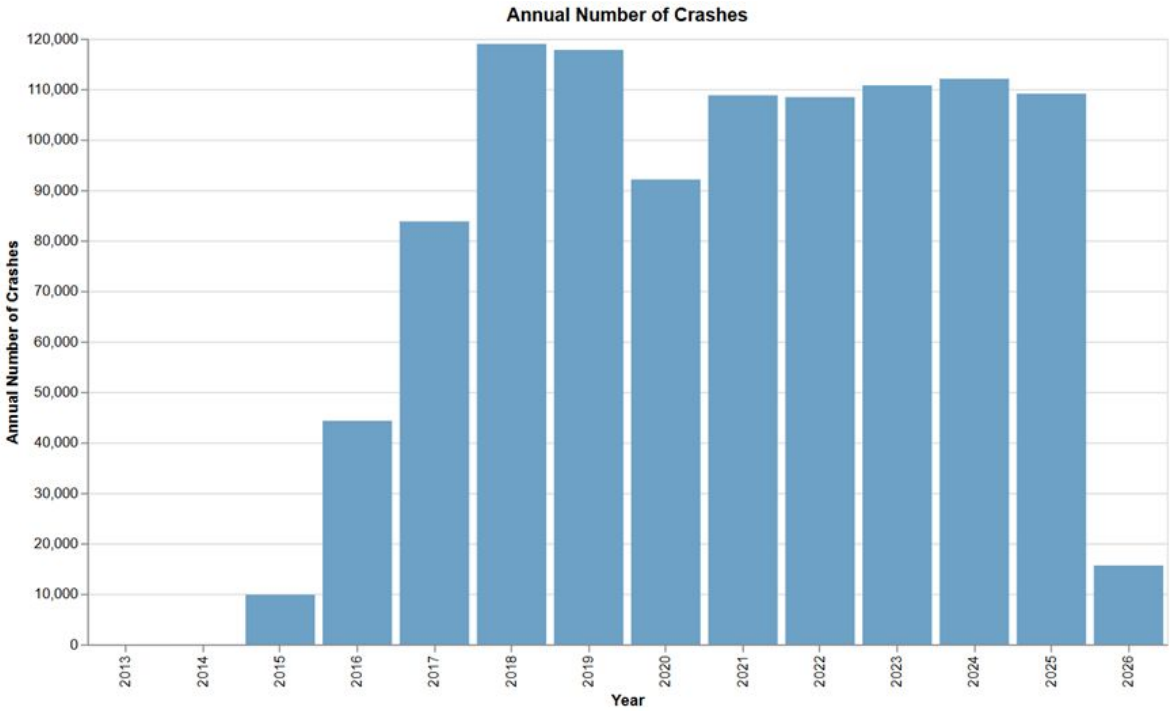
Research Framework:

- Descriptive analysis
- Event-study
- Dashboard

Overview of Traffic Crashes in Chicago

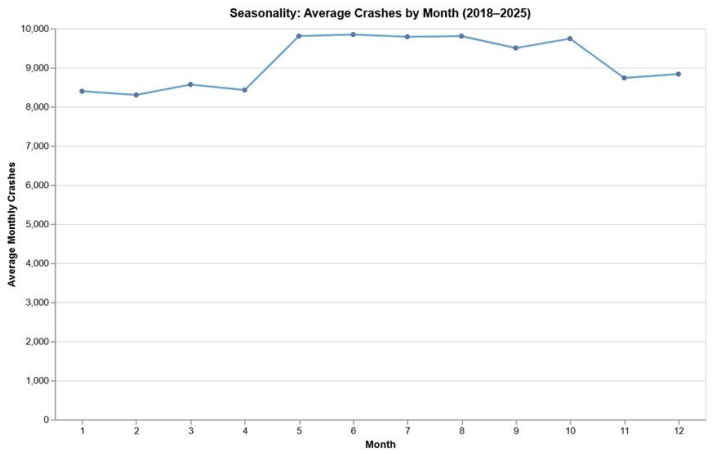
Selection of the Analysis Period

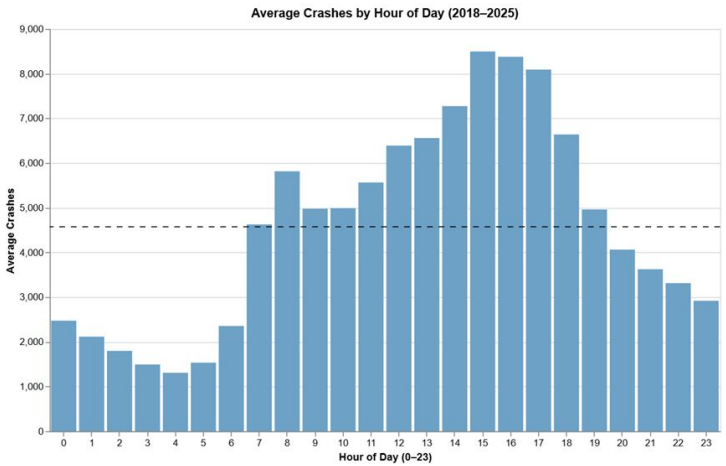


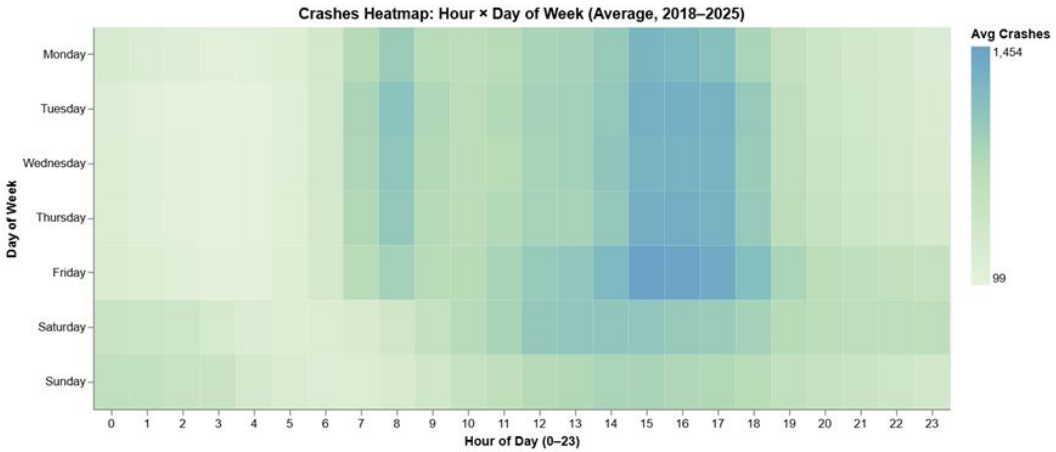


We restrict our analysis to data from 2018 onward.

Temporal Patterns of Traffic Crashes



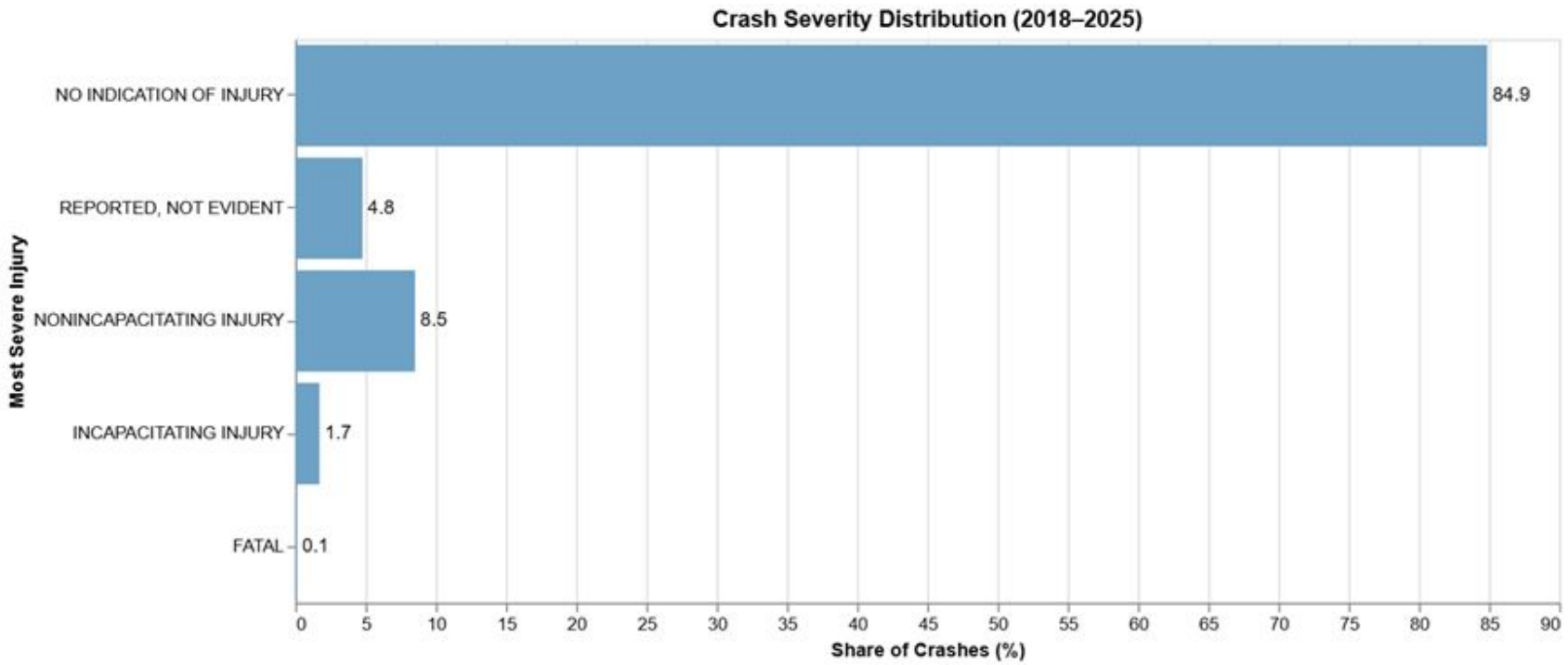




- **Monthly: Crashes increase in late spring and summer.**
- **Hourly: Crashes peak during afternoon rush hours.**
- **Weekly × Hourly: Weekday afternoons, especially Fridays, show the highest concentration.**

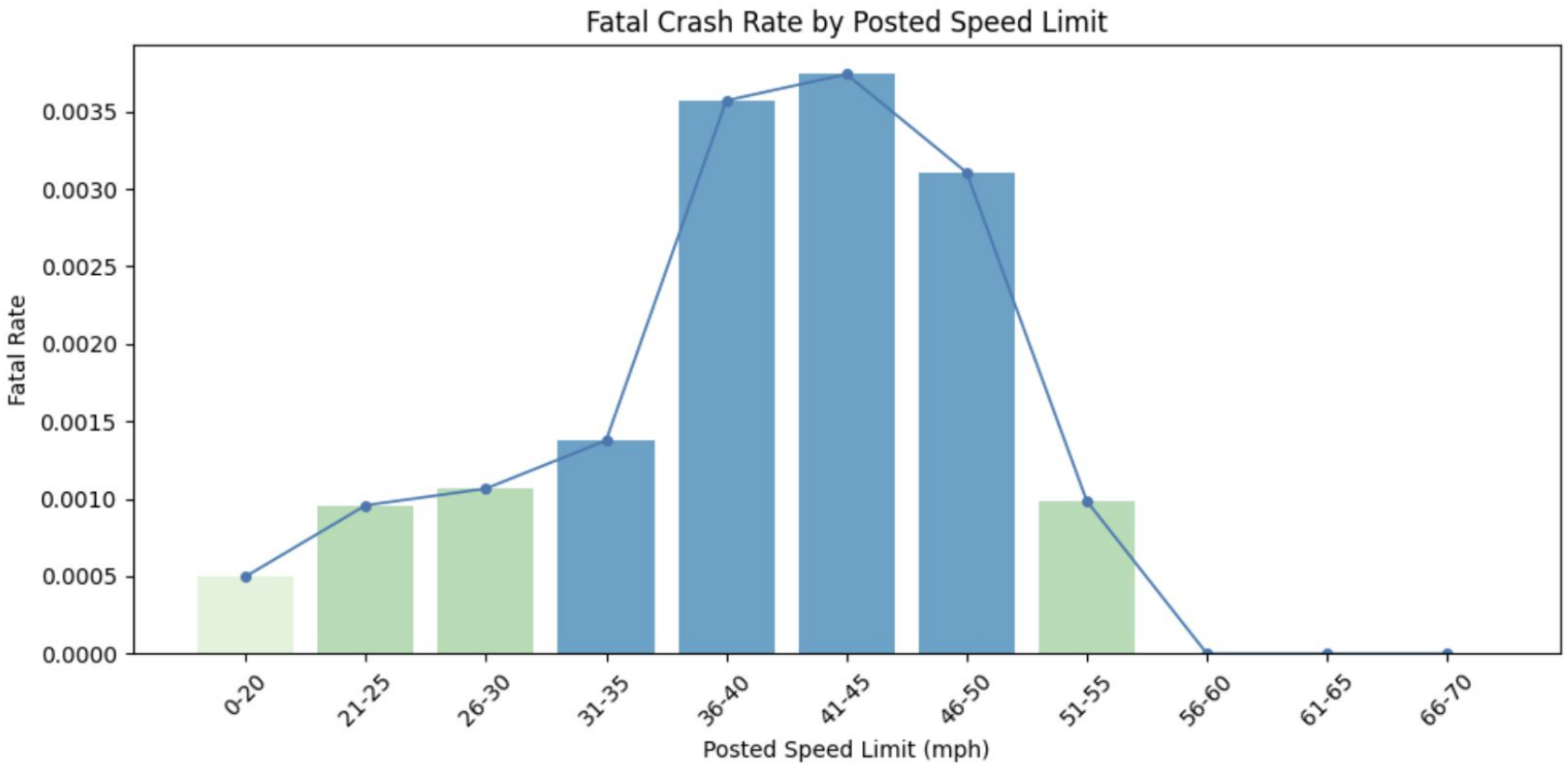
Crash frequency is largely driven by traffic exposure and mobility patterns.

Distribution of Crash Severity



- ~85% involve no reported injury
- Severe injuries are relatively rare
- Fatal crashes represent a very small share

Fatal Crash Rate by Posted Speed Limit

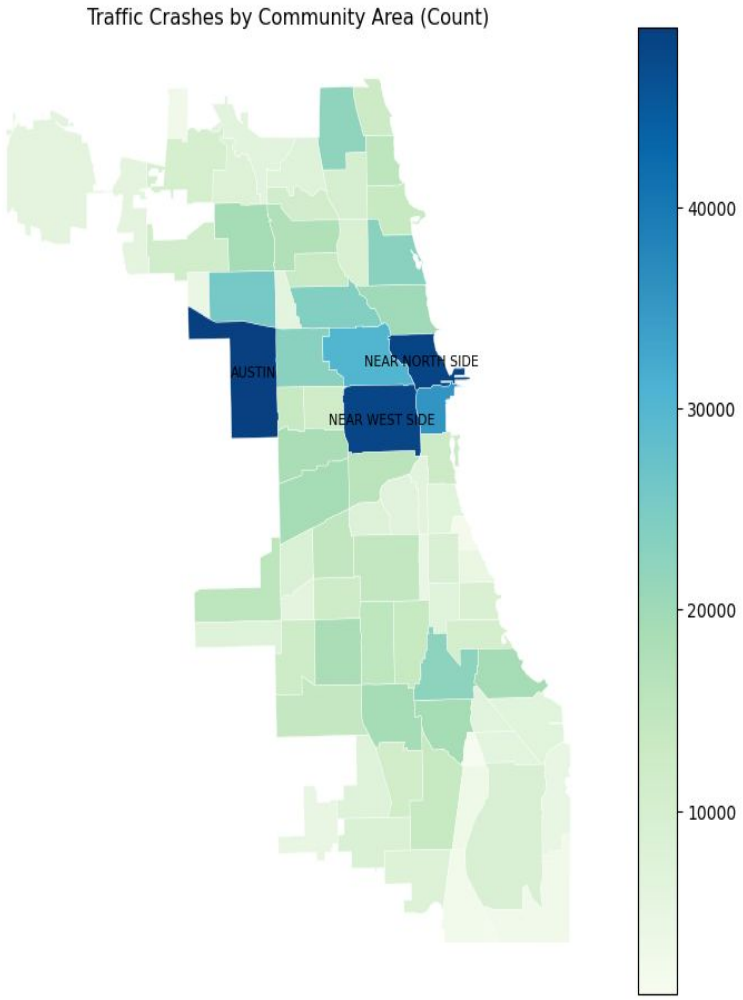


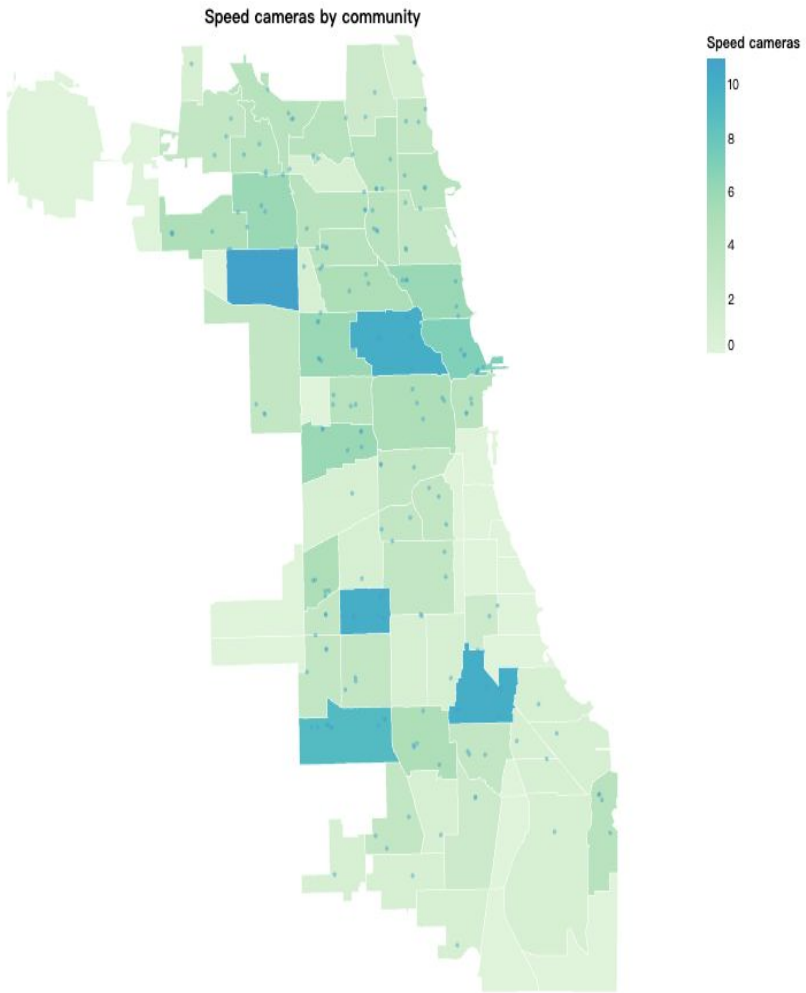
- Fatal risk increases with higher speed limits
- Peak fatal rate around 40–50 mph
- Slight decline at very high speeds

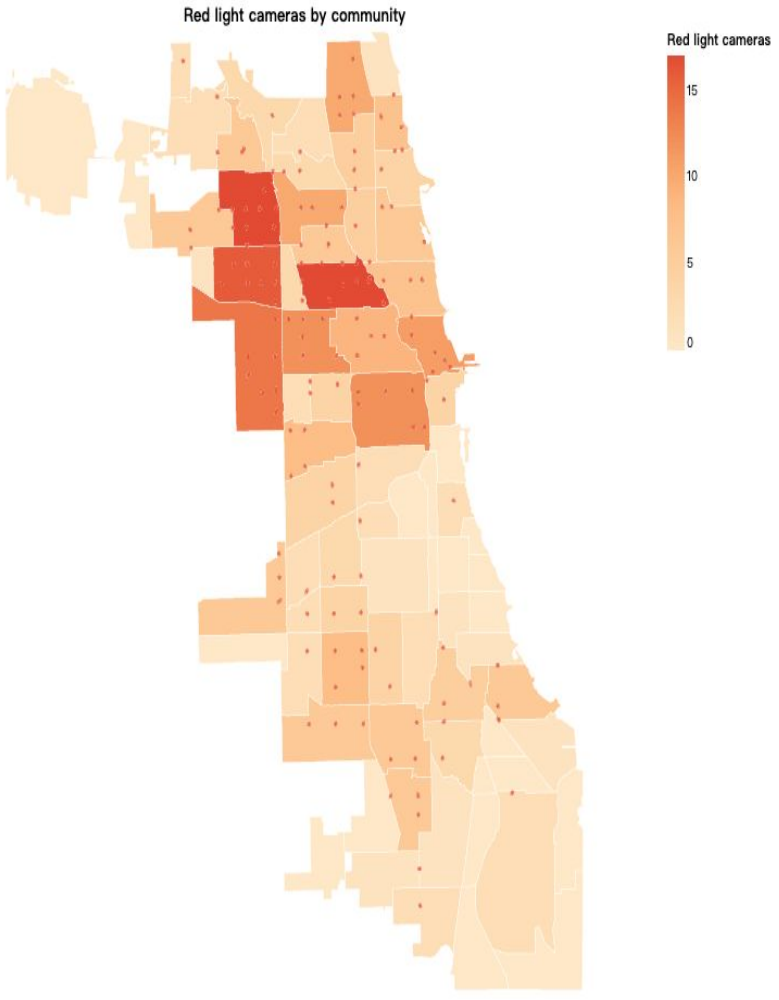
Speed regulation plays a critical safety role

Spatial Distribution of Transportation Camera Deployment in Chicago

Geographic Distribution of Crashes and Cameras in Chicago





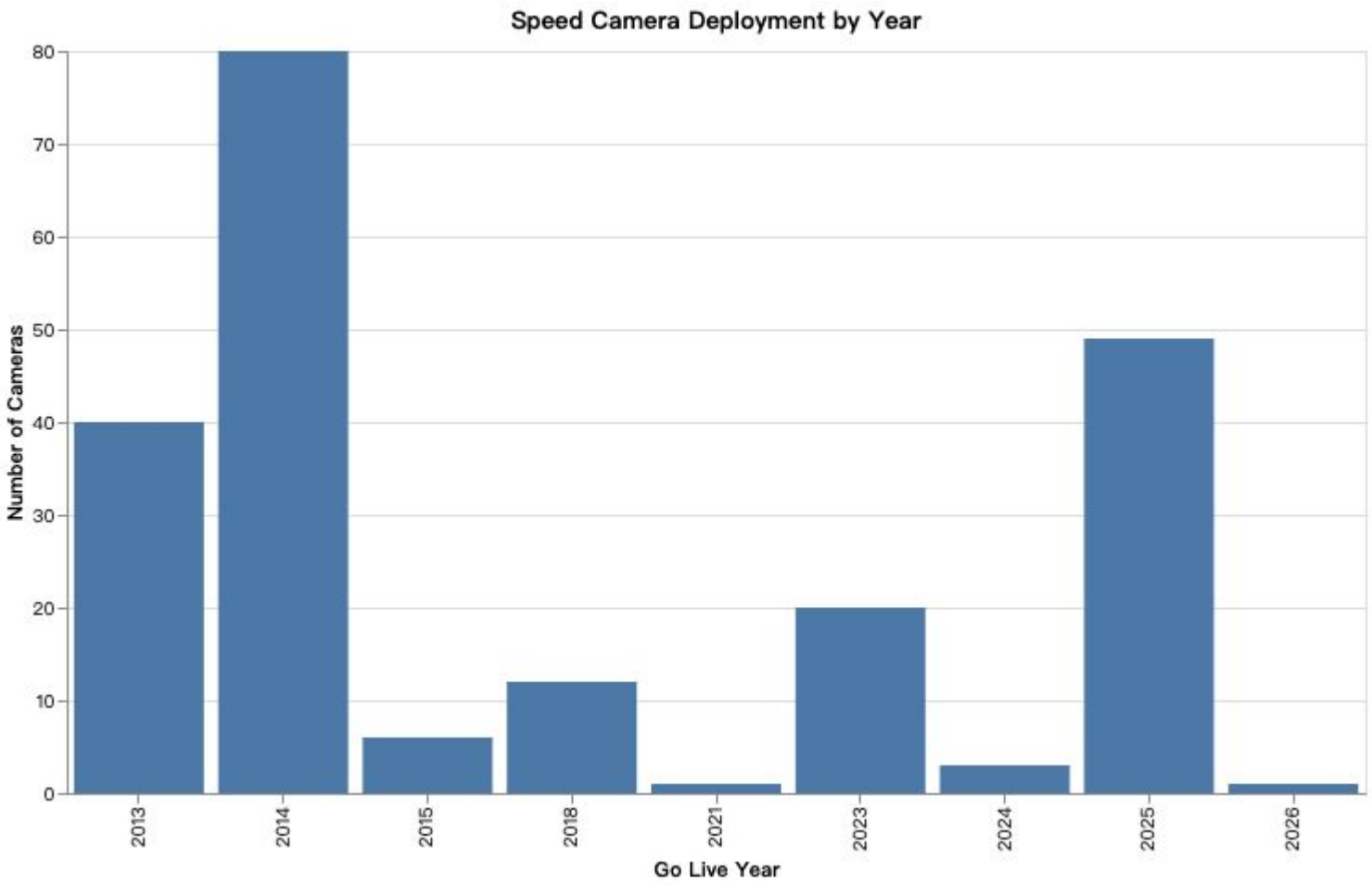


Event Study

Timing of Speed Camera Deployment

Two major deployment waves:

- The largest expansion occurred in 2013-2014
- A second expansion began around 2023



Event Study Design - Speed Camera

Why Event Study?

- Speed cameras are installed at different times
- We align crashes relative to each installation date

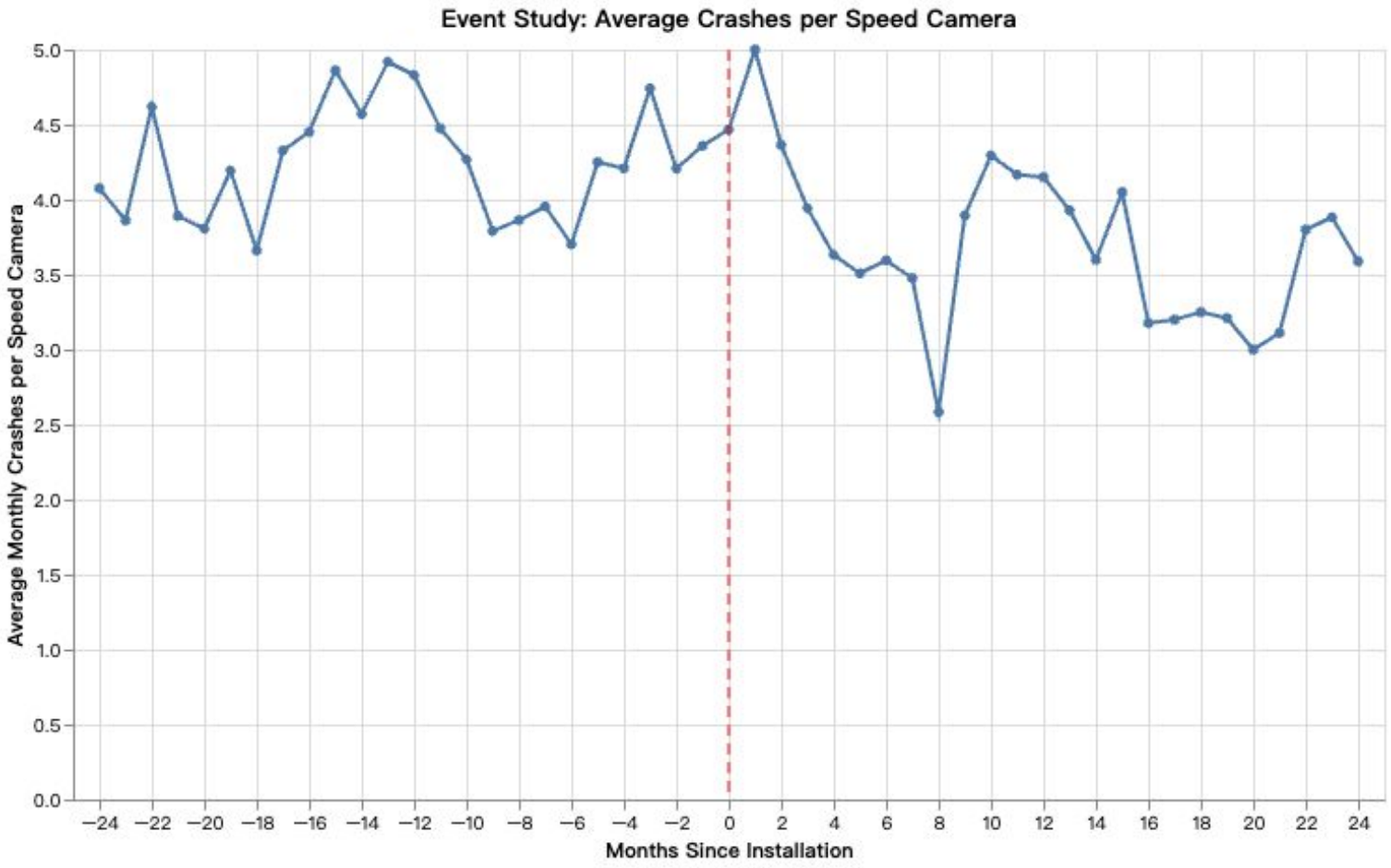
Sample Restriction

- We restrict our analysis to cameras installed in the second period (after 2023) because our crash data only began in 2013.
- Cameras installed earlier would not have an observable pre-treatment period.

Treatment Definition

- To define treatment exposure, we construct a 200-meter buffer around each camera, so the safety effect is highly localized.

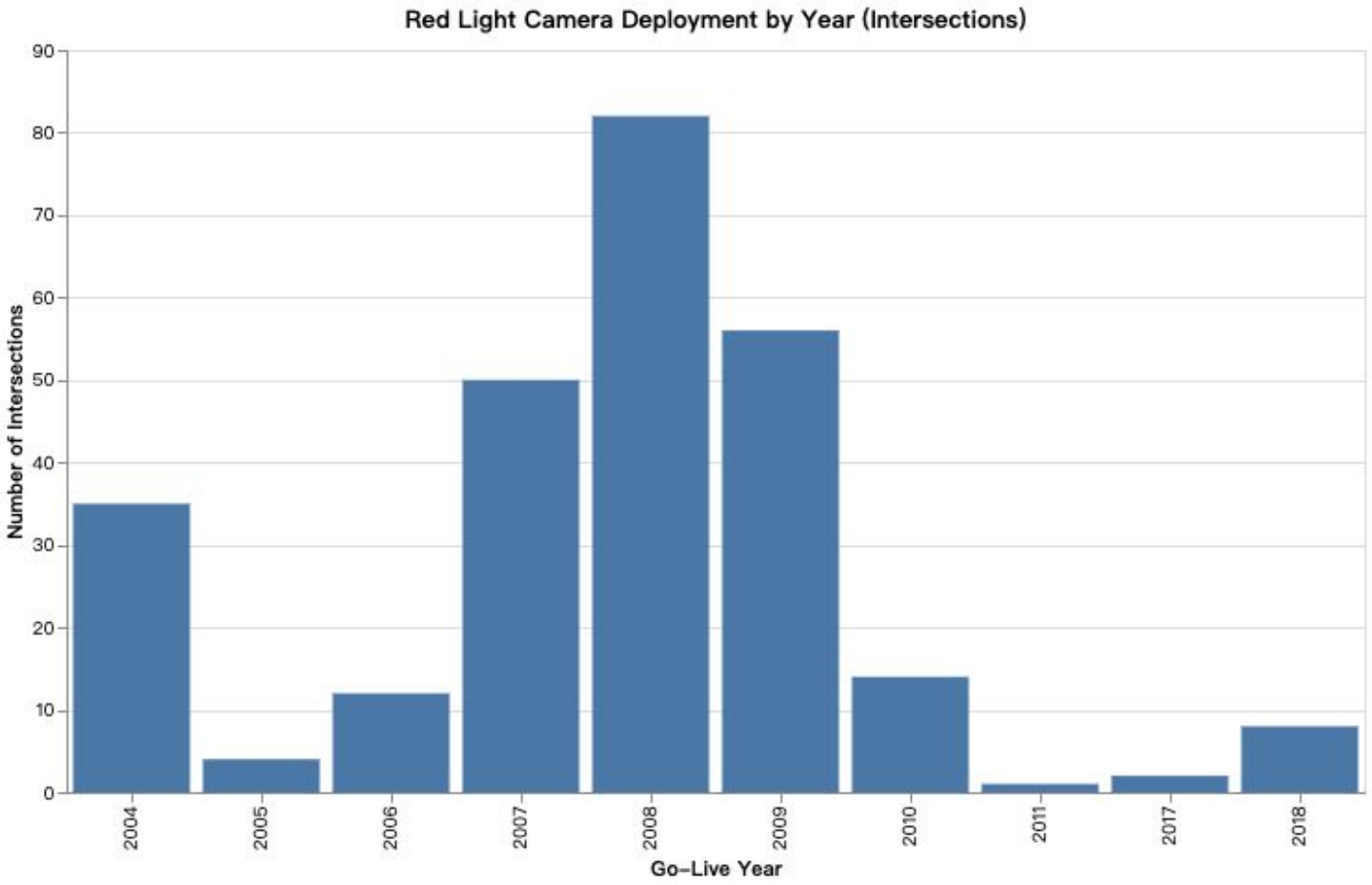
Key Findings - Speed Camera



- Pre-treatment period: crashes remain stable, suggesting no evidence of anticipatory decline or pre-trend violation
- After installation: approximately 15-25% reduction in crash frequency

Timing of Red Light Camera Deployment

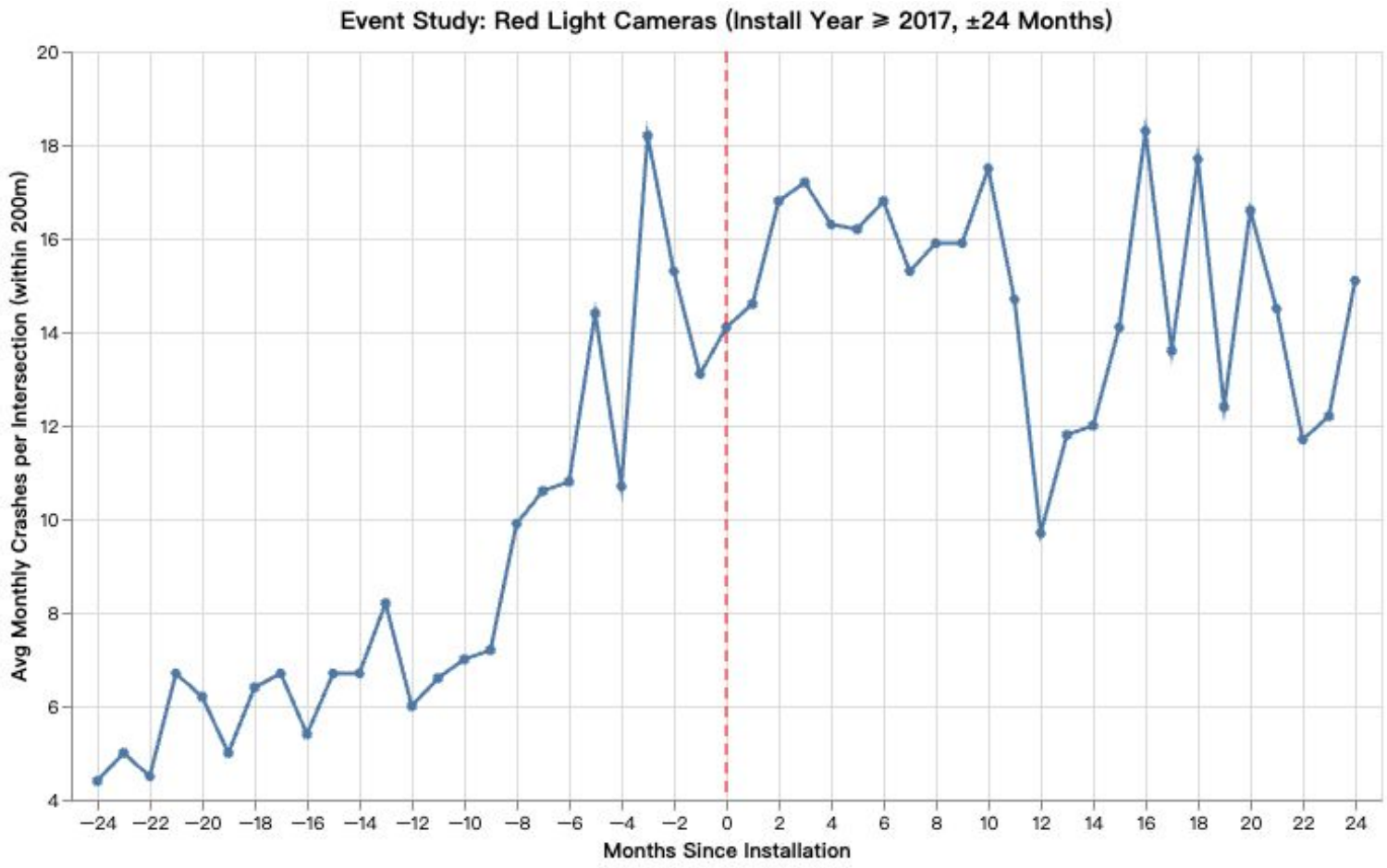
- The majority of red light cameras were installed before 2010
- A subset was deployed through 2017-2018



Event Study Design - Red Light Camera

- We apply the same event-study framework as in the speed camera analysis
- Crashes are aligned relative to each intersection's go-live date
- Treatment defined as crashes within 200 meters
- Due to data limitations, we restrict the sample to installations after 2017 to ensure a pre-treatment period

Key Findings - Red Light Camera



- Pre-treatment period: strong upward trend, which may suggest that intersections with increasing crash risk are more likely to receive red light cameras.
- After installation: crash levels appear to stabilize rather than continue increasing

Dashboard

Dashboard

https://final-project-chicago-crashes-g9srdmczqbxunyfhgnttyg.streamlit.app/

Connection to Policy

Connection to Policy

How results can inform decisions:

- Expand cameras in high-crash, high-risk areas
- Use data to guide future deployment
- Focus enforcement during peak traffic hours

Questions?

Thank you